

Load (N)	Extension (mm)	
5	1.2	$y = mx + c$ $m = \frac{n \sum xy - \sum x \sum y}{n \sum (x^2) - (\sum x)^2}$
10	2.5	
15	3.7	
20	4.9	
25	6.1	

x	y	xy	x ²
5	1.2	6	25
10	2.5	25	100
15	3.7	55.5	225
20	4.9	98	400
25	6.1	152.5	625
$\sum x$ = 75	$\sum y$ = 18.4	$\sum xy$ = 337	$\sum x^2$ = 1375

$$m = \frac{5(337) - (75 \times 18.4)}{5(1375) - 5625} = 0.244$$

$$c = \frac{\sum y - m \sum x}{n} = \frac{18.4 - 0.244 \times 75}{5} = 0.02$$

For Load $x = 55$

$$\begin{aligned} \text{Extension } y &= 0.244 \times 55 + 0.02 \\ &= 13.44 \end{aligned}$$